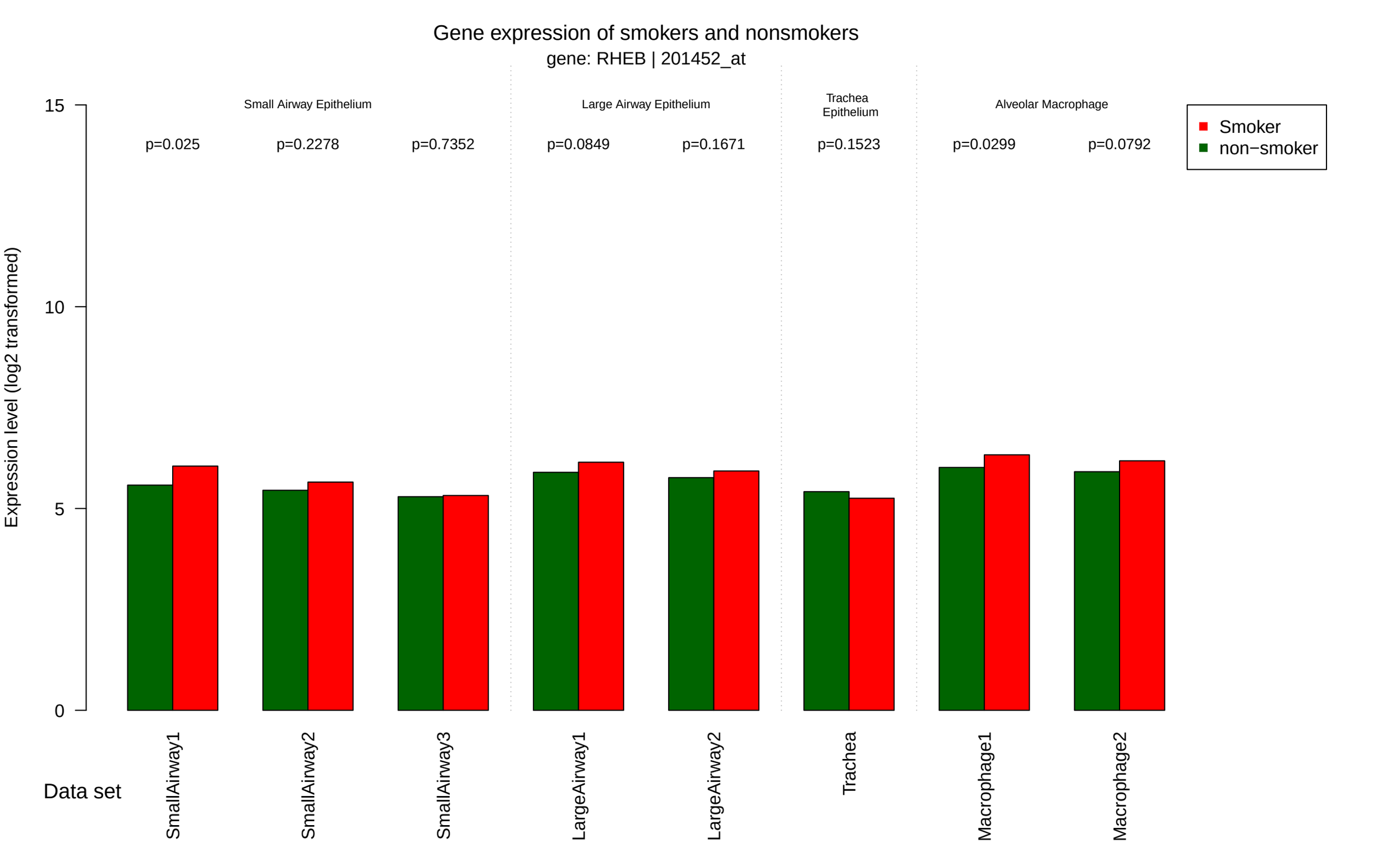
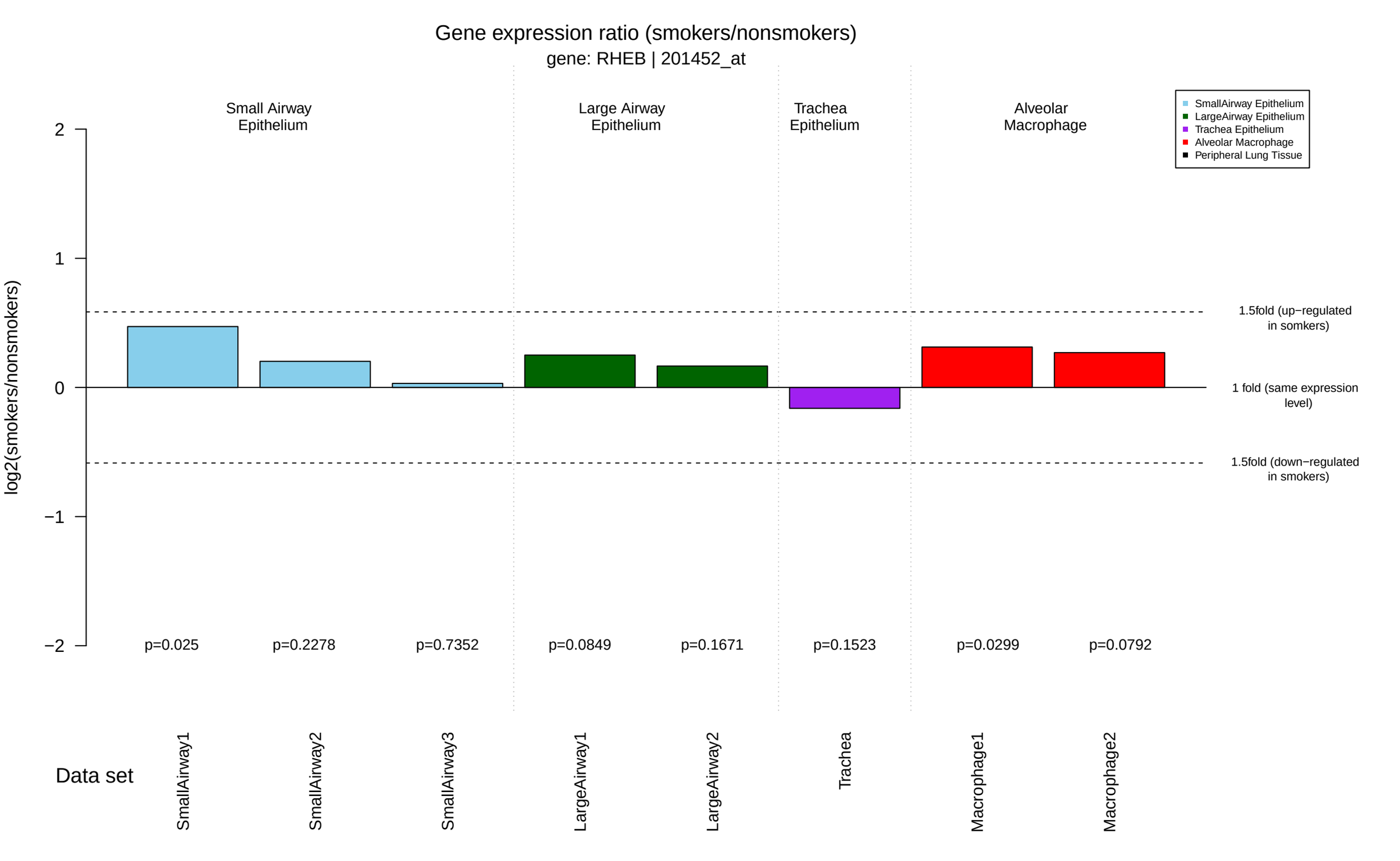
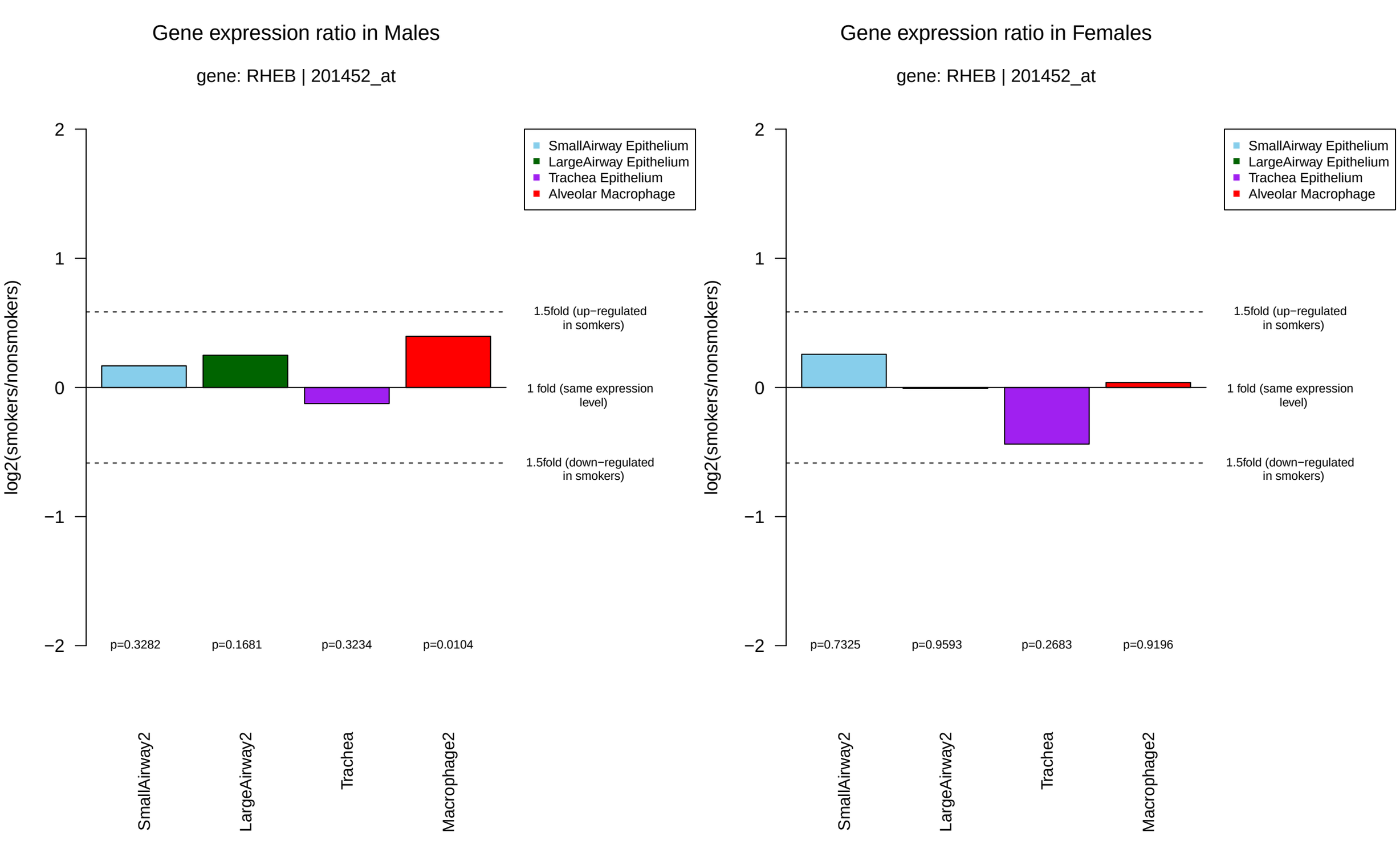
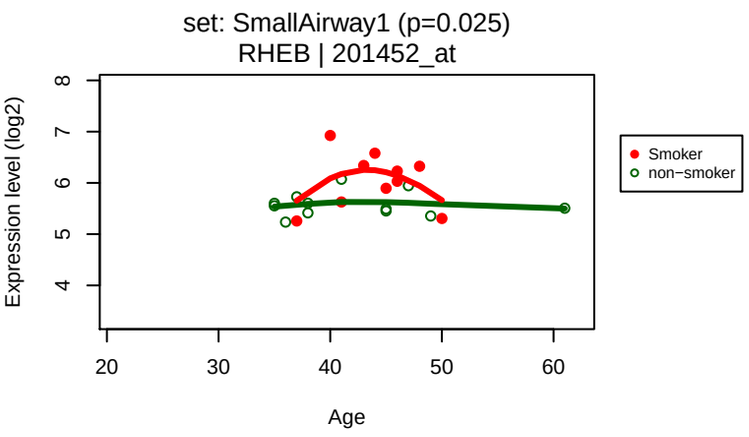
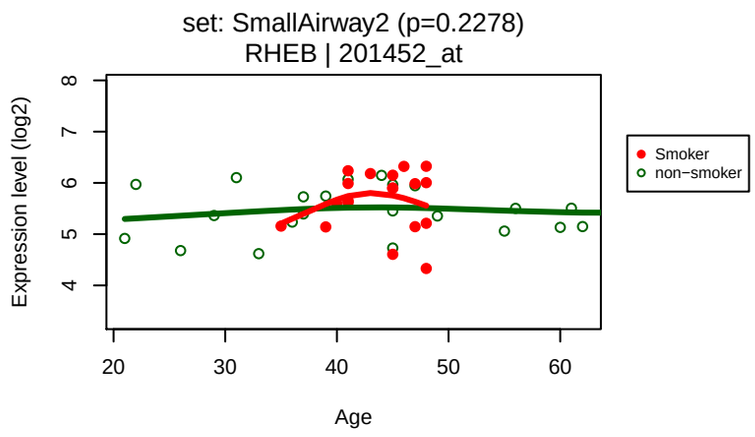
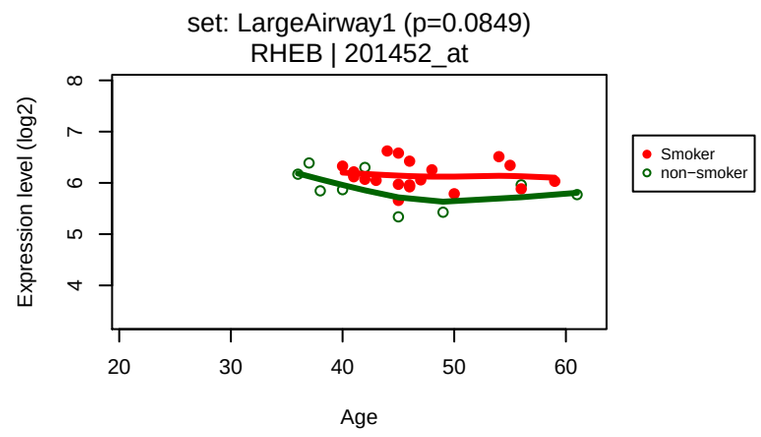

Gene expression of smokers and nonsmokers
gene: RHEB | 201452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.025
p=0.2278
p=0.7352
p=0.0849
p=0.1671
p=0.1523
p=0.0299
p=0.0792
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

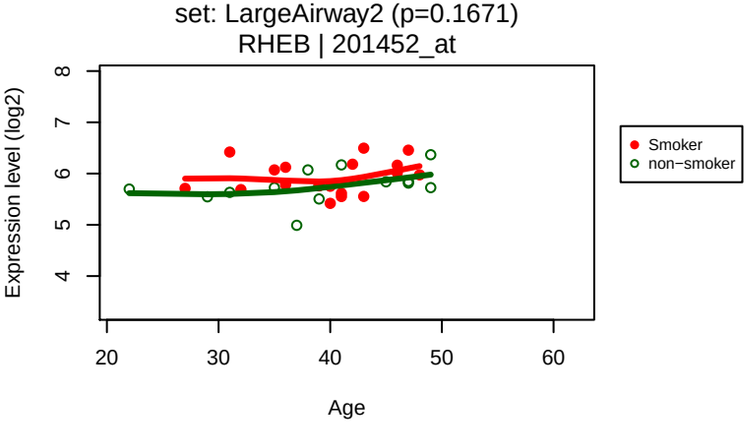

Gene expression ratio (smokers/nonsmokers)
gene: RHEB | 201452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.025
p=0.2278
p=0.7352
p=0.0849
p=0.1671
p=0.1523
p=0.0299
p=0.0792
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

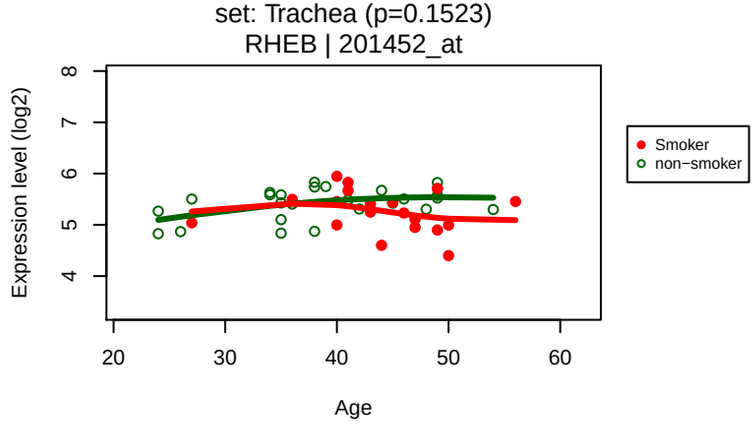

Gene expression ratio in Males
gene: RHEB | 201452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3282
p=0.1681
p=0.3234
p=0.0104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHEB | 201452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7325
p=0.9593
p=0.2683
p=0.9196
SmallAirway2
LargeAirway2
Trachea
Macrophage2

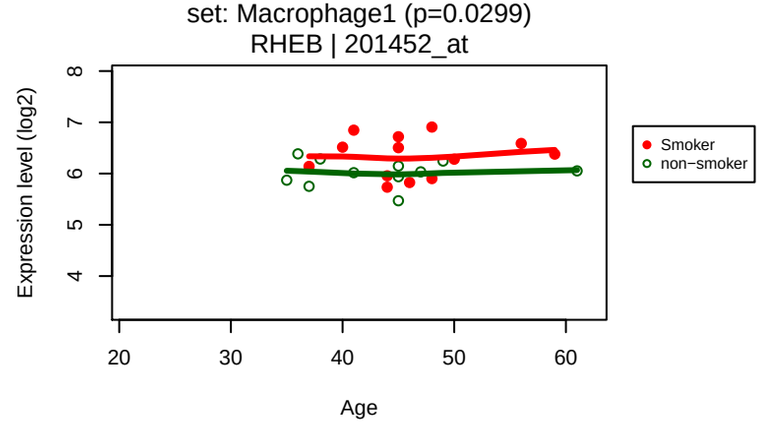

set: SmallAirway1 (p=0.025)
RHEB | 201452_at
Expression level (log2)
Age
Smoker
non−smoker

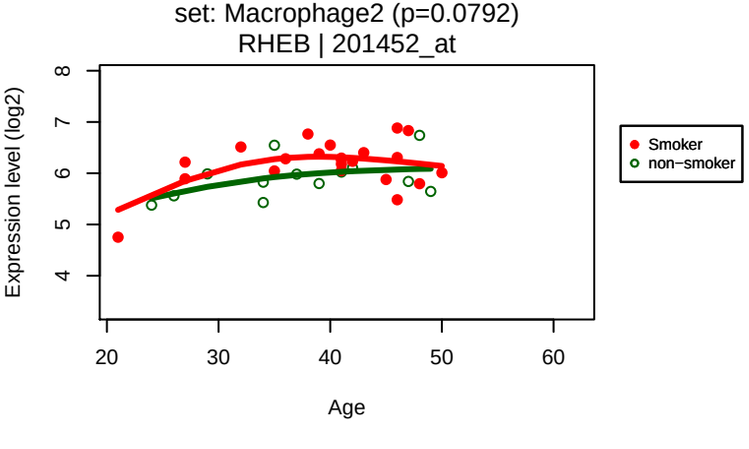

set: SmallAirway2 (p=0.2278)
RHEB | 201452_at
Expression level (log2)
Age
Smoker
non−smoker

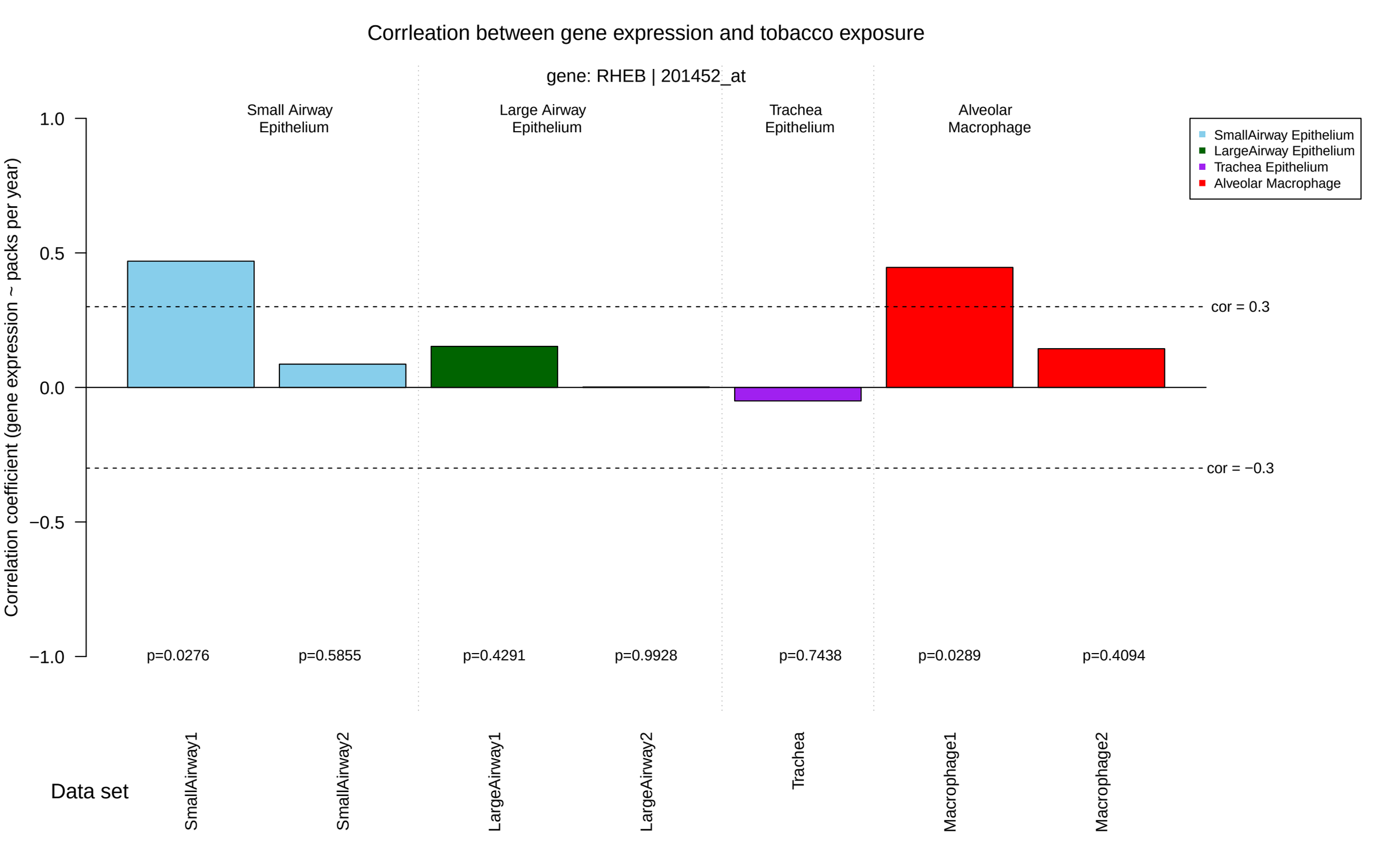

set: LargeAirway1 (p=0.0849)
RHEB | 201452_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1671)
RHEB | 201452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1523)
RHEB | 201452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0299)
RHEB | 201452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0792)
RHEB | 201452_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHEB | 201452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0276
p=0.5855
p=0.4291
p=0.9928
p=0.7438
p=0.0289
p=0.4094
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2